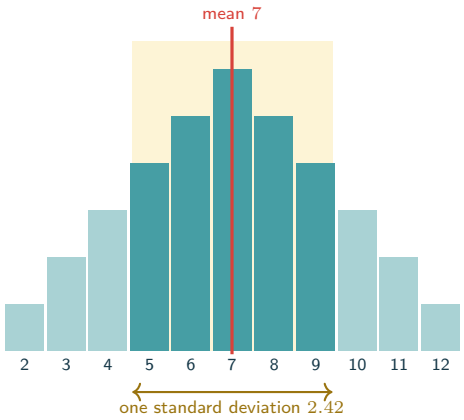
PROBABILITY LADDER 5 OF 9

Random variables, expectation and variance

A random variable attaches a number to each outcome, and its mean and spread summarise the whole distribution.

1 / THE RULE AND THE PICTURE

The mean is the weighted average, and the variance is the average squared distance from it



the sum of two dice, with bar heights proportional to $P(X = s)$

For a random variable X with probabilities $p(x)$,

$$E[X] = \sum x\,p(x), \quad \operatorname{Var} X = E[X^2] - \big(E[X]\big)^2$$

For the sum of two dice $E = 7$, $\operatorname{Var} = \frac{35}{6} = 5.83$ and the standard deviation is $\sqrt{35/6} = 2.42$.

2 / ONE DIE

The mean of a fair die is 3.5 though no roll shows it

With $p = \frac{1}{6}$ each, $E[X] = \dfrac{1+2+\cdots+6}{6} = \frac{7}{2}$ and $E[X^2] = \dfrac{1+4+9+16+25+36}{6} = \frac{91}{6}$, so

$$\operatorname{Var} X = \tfrac{91}{6} - \left(\tfrac{7}{2}\right)^2 = \tfrac{35}{12} = 2.92, \qquad \sigma = 1.71.$$

The mean is the balance point of the distribution, and the standard deviation measures how far a typical roll sits from it.

3 / COMBINING VARIABLES / LINEARITY

Means always add, and variances add for independent variables

rule	statement	example
mean of a sum	$E[X+Y] = E[X] + E[Y]$	two dice: $\frac{7}{2} + \frac{7}{2} = 7$
variance of a sum, independent	$\operatorname{Var}(X+Y) = \operatorname{Var}X + \operatorname{Var}Y$	$\frac{35}{12} + \frac{35}{12} = \frac{35}{6}$
scaling and shifting	$E[aX+b] = aE[X] + b$	double and add 1: $2 \cdot 3.5 + 1 = 8$
variance of a scaling	$\operatorname{Var}(aX+b) = a^2 \operatorname{Var}X$	doubling a die: $4 \cdot \frac{35}{12}$

The mean rule needs no independence at all, which is why it is so useful, while the standard deviation of the sum is $\sqrt{2}$ times that of one die and not twice as large.

4 / EXPECTED VALUE IN GAMES

A fair price for a game equals its expected winnings

game	expected winnings	meaning
roll a die, win its value in pounds	$\frac{7}{2}$	fair price £3.50
bet £1 on red at American roulette	$\frac{18}{38} - \frac{20}{38} = -\frac{1}{19}$	loses 5.3p on average
lottery jackpot £1 000 000 from $\binom{49}{6}$ tickets	$\frac{10^6}{13\,983\,816}$	7.2p from the jackpot alone

A negative expectation does not predict any single play, but over many plays the average result converges to it, which is how casinos make their money.

5 / INDICATOR VARIABLES / COUNTING WITHOUT LISTING

Write a count as a sum of 0-1 variables and add their means

The number of aces in a five-card hand is the sum of five indicators, each with mean $\frac{4}{52}$, so the expected number is $5 \cdot \frac{4}{52} = \frac{5}{13} = 0.385$, even though the cards are dependent.

The number of fixed points of a random permutation of n objects has mean $n \cdot \frac{1}{n} = 1$ for every n, and a simulation over all 120 permutations of five objects confirms the average of exactly 1. The expected number of heads in ten tosses is $10 \cdot \frac{1}{2} = 5$.

ASIDE / MEAN IS NOT TYPICAL

A mean can be a value that never occurs, such as 3.5 for a die, so it describes the long-run average and not a single outcome.

LINKS TO OTHER SHEETS

The dice sums come from *Sample spaces and equally likely outcomes*; the next sheet is *The binomial distribution*, which has mean np.

Convention. $E[X]$ is the expected value; $\operatorname{Var} X$ the variance; $\sigma = \sqrt{\operatorname{Var} X}$ the standard deviation; dice are fair and independent.

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